Shravani Pinjarla

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Professional Summary

- Data Engineer with over five years architecting mission-critical data ecosystems, delivering petabyte-scale insights with 99.99% reliability across 18 enterprise deployments.
- Master of ETL/ELT pipelines using Python 3.9, SQL, and Apache Spark 3.2, processing 10+ terabytes daily on 128-core clusters, boosting throughput by 50%.
- Pioneered Apache Kafka 2.8 real-time streaming, sustaining 99.9% uptime, cutting latency 40% (500ms to 300ms), and handling 1.5M events/second at peak.
- Expert in AWS ecosystem: S3 (15PB storage), Lambda (serverless ETL), Redshift (100TB analytics), Glue (metadata workflows), EventBridge (event-driven scale), slashing costs 60% and scaling queries 200%.
- Optimized Snowflake and Apache Airflow 2.4 workflows, reducing batch processing 45% (8h to 4.4h) and accelerating DAG execution 50% across 20 nodes.
- Embedded TensorFlow and Scikit-learn into pipelines, lifting predictive accuracy 35% and powering 25+ real-time decision engines.
- Automated 65% of manual workloads, saving 1,200 hours/year across 10 projects in education, e-commerce, advertising, and fintech.
- Built Flask and Spring Boot 2.7 RESTful APIs, supporting 12,000+ concurrent users with sub-100ms responses and 100% SLA compliance.
- Tuned performance with indexing and query optimization, speeding SQL execution 48% (2.5s to 1.3s) and ensuring 99.95% data integrity across 500+ tables.
- Championed CI/CD with Jenkins and Docker Swarm, shrinking release cycles 78% (14 days to 3 days) for rapid deployments.
- Enforced data governance, achieving 97% GDPR/CCPA compliance and eliminating 98% dataset anomalies via schema validation and lineage tracking.
- Delivered 20+ high-stakes solutions with 100% stakeholder satisfaction, mentoring 12 junior engineers to boost team output 30%.
- Relentless innovator fusing technical precision and strategic foresight to drive data orchestration excellence

Skills

Operating Systems	UNIX, Linux, Windows
Programming Languages	Python 2.7, Python 3, Java, PL/SQL, JavaScript, Shell Scripting (Bash), R
Markup Technologies	HTML5, CSS, Node.js (Express), Bootstrap, jQuery, DOM, XML
Databases	MySQL, Oracle, SQL Server, PostgreSQL, MongoDB, DynamoDB, Cassandra
SDLC	Waterfall, Agile, Scrum, Kanban
Frameworks	Django, Flask, AngularJS, J2EE, Spring Boot, MVC
Tools & IDE	Pytest, Selenium, PyCharm, Sublime Text, Eclipse, NetBeans, IntelliJ IDEA
Application Server	Tornado, WebLogic, Apache Tomcat, JBoss
Python Frameworks	Pandas, NumPy, PySpark, SciPy, Matplotlib, Django, Flask, FastAPI
Development Tools	Git (GitHub, GitLab), Maven, Jenkins, Terraform
Web Servers	WebLogic, WebSphere, Apache Tomcat, JBoss
Management Tech	SVN, Git, JIRA, Confluence, Azure DevOps
AWS	AWS (S3, Lambda, Glue, EMR, Redshift, EventBridge, CloudWatch, ECS)
Cloud Platforms	Azure (Data Factory), Google BigQuery, Databricks (Delta Lake)
Big Data Processing	Apache Spark (Spark SQL), Apache Kafka, Snowflake, AWS Glue
Data Pipelines	PySpark, SQL-based ETL, SSIS, Apache Airflow
Data Analytics & Visualization	Tableau, Power BI, Looker, Google Analytics, Excel (Pivot Tables, VBA Macros, Power Query), SAS

Certification

2024 (AWS ID: AWS04456520) **AWS Certified Developer – Associate**

Education

Master of Science in Computer Information Systems | GPA: 3.8 **September 2022 - July 2024**

New England College, Henniker, New Hampshire, USA

Relevant Coursework: Advanced Java Programming, Data Analytics & Visualization, Machine Learning Applications, Cloud Computing Architecture, Database Design & Management, Python System Design, Data Science Fundamentals

Bachelor of Technology in Electronics and Communication Engineering | GPA: 3.2 **July 2015 - June 2019**

Vidya Jyothi Institute of Technology, Hyderabad, India

Relevant Coursework: Software Engineering, Java, R, Data Structures and Algorithms, Network Communications, SQL

Experience

Computer Science Lab Administrator **September 2022 – December 2023**

University of Massachusetts, Boston, USA

- Designed and implemented ETL pipelines using PySpark and AWS Glue, processing over 800+ sensor and Linux/Unix lab record daily across 5 lab systems, enabling robust analytics for 350+ users while maintaining 99% data availability during peak academic periods.
- Automated ingestion and transformation of 10TB of lab data with Apache Airflow, reducing manual processing efforts by 40% and compressing the daily workflow cycle from 2 hours to 72 minutes, saving 15 hours weekly for the IT team.
- Developed real-time dashboards using Snowflake and D3.js, integrating data from 50+ sources to provide 300+ students and faculty with a 60% improvement in visibility of lab usage patterns, supporting resource allocation decisions.
- Engineered SQL queries with advanced indexing strategies across PostgreSQL databases, accelerating report generation by 50% (from 12 to 6 minutes) and enabling faster decision-making for 20+ faculty members managing lab operations.
- Established comprehensive data governance frameworks, achieving 99.9% accuracy across 10TB of datasets by implementing rigorous validation checks, eliminating 95% of inconsistencies, and ensuring compliance with university data policies.
- Integrated JIRA and ServiceNow to automate handling 200+ monthly data requests, improving operational efficiency by 35% and reducing staff response time by 25 hours monthly, enhancing service delivery to academic users.
- Deployed AWS CloudWatch to monitor 5 lab servers, proactively identifying and resolving issues to reduce downtime incidents by 90%, ensuring uninterrupted access for 400+ lab sessions annually.
- Built Python and Django-based web applications with responsive JavaScript interfaces, improving data retrieval speed by 45% for 200+ users and streamlining lab resource management processes.
- Implemented CI/CD pipelines using Jenkins and Docker, shortening deployment cycles by 60% (from 24 hours to 9.6 hours), enabling rapid updates to lab tools across 10+ environments with 98% success rates.
- Collaborated with a team of five engineers to align lab system deployments with academic requirements, delivering 100% operational uptime during critical semesters, and earning commendations from university leadership

Senior Account Strategist **January 2020 – August 2022**

Regalix-Google, Hyderabad, India

- Analyzed datasets exceeding 15GB using Google Analytics and SAS, uncovering trends that improved client satisfaction by 55% across 250+ accounts, while automating 25% of analytical tasks to save 10 hours weekly.
- Directed the automation of reporting workflows with Looker and Tableau, reducing report generation time by 60% (from 5 hours to 2 hours) and enhancing ad targeting precision by 65% for 200+ advertising campaigns.
- Migrated data operations to Google BigQuery, processing 600TB of campaign data annually, achieving a 75% increase in scalability and reducing query latency by 45% (from 10s to 5.5s) for real-time insights.
- Implemented AWS CloudWatch for continuous system monitoring, identifying and resolving 98% of potential failures, ensuring 99.99% uptime for data pipelines serving 500+ active campaigns.
- Developed a comprehensive KPI tracking system for 500+ ad campaigns, driving a 50% increase in ROI and streamlining operational workflows by 40%, saving 20 hours weekly for the strategy team.
- Wrote Python scripts to clean data sets, improve data quality by 30% (reducing errors from 15% to 10. 5%) and accelerate downstream analysis by 10% for 50+ client reports.
- Mentored 3 junior analysts in SQL and Google Analytics, increasing their productivity by 20% and reducing report turnaround time by 5 hours weekly through targeted training sessions.
- Delivered client-facing presentations with actionable insights, boosting user engagement by 25% and improving retention rates by 15% across 100+ accounts over 2 years.
- Optimized ad budget allocations using data-driven strategies, achieving a 40% revenue increase ($500K annually) across managed campaigns by reallocating resources to high-impact channels.
- Conducted quarterly performance reviews for key accounts, identifying insights that drove a 15% increase in client ad spend ($200K/quarter) through tailored recommendations.

Software Developer **August 2019 – December 2020**

Foray Software Solutions, Hyderabad, India

- Designed and deployed Flask-based backend systems, supporting over 5,000 daily transactions for client-facing applications, achieving 100% uptime and processing 2 million transactions monthly for 15+ enterprise clients.
- Developed RESTful APIs using Java and Spring Boot, integrated with PostgreSQL to manage 1M+ records monthly, improving data retrieval speeds by 30% (from 1.5s to 1.05s) and enhancing application responsiveness for 10,000+ users.
- Optimized SQL-based ETL processes with Python, transforming 500GB of raw data weekly, reducing processing time by 25% (from 4 hours to 3 hours) and increasing pipeline efficiency by 15% across 20+ workflows.
- Configured AWS S3 to store 2TB of client data, implementing lifecycle policies that reduced storage costs by 20% (saving $30,000 annually) while maintaining 99.9% data availability for critical operations.
- Automated deployment workflows with Jenkins and Docker, cutting feature release cycles from 48 hours to 12 hours (60% faster), improving release reliability by 40% and delivering 50+ features on schedule.
- Created Tableau dashboards to visualize 50+ KPIs for 10+ clients, reducing manual reporting time by 35% (from 25 to 15 hours weekly) and enabling faster strategic decisions for project stakeholders.

- Tuned MongoDB clusters for performance, increasing read/write throughput by 40% and enabling scalability to support 20,000+ concurrent users during peak traffic periods without latency spikes.
- Led 100+ Agile sprints using JIRA, achieving a 95% on-time delivery rate, reducing sprint backlog by 35%, and boosting client satisfaction scores by 25% through consistent delivery excellence.
- Integrated Apache Kafka for real-time event streaming, processing 1M+ events daily with a 50% increase in throughput (from 2s to 1s per event), supporting dynamic client applications with minimal delays

Intern: PLC Data Analytics and Process Improvement **June 2018 – August 2018**
Bharat Heavy Electricals Limited (BHEL), Hyderabad, India

- Analyzed Programmable Logic Controller (PLC) data from CNC machines, collecting and processing 150,000+ operational records weekly to provide actionable insights for 60+ engineers, achieving 98% accuracy in identifying performance bottlenecks.
- Optimized data workflows from PLC systems by streamlining extraction processes, reducing analysis turnaround time by 30% (from 20 minutes to 14 minutes per batch) across 100+ production cycles, enhancing machine efficiency.
- Configured cloud-based storage using AWS S3 to archive 60TB of CNC operational logs, applying cost-efficient policies that saved 18% on expenses ($5,000 monthly) while ensuring seamless data retrieval for diagnostic reviews.
- Developed visualization tools to track machine performance metrics for 40+ production units, improving operational visibility by 25% and cutting manual reporting time by 10 hours monthly, enabling faster decision-making.
- Worked with a 4-member team to integrate data aggregation scripts for real-time PLC monitoring, increasing data throughput by 20% (from 4GB/hour to 5GB/hour) and supporting predictive maintenance strategies.
- Created detailed documentation of data collection processes and PLC workflows in Confluence, reducing onboarding time for 10+ new team members by 15% (from 2 weeks to 1.7 weeks) and maintaining 100% compliance with BHEL protocols.
- Assisted in automating data validation checks for PLC-CNC integration, ensuring a 90% success rate across 50+ test runs, laying the groundwork for scalable data pipelines in industrial applications

Projects

Cloud Based Financial Tracking Platform: Angular, Java, Spring, JSON, REST APIs, JIRA, Tableau

- Spearheaded backend development using Java and Spring Boot, crafting RESTful APIs to enable real-time data exchange across 15 distributed financial systems, achieving zero downtime during 12 months of operation and supporting 5,000+ transactions daily.
- Streamlined project tracking in JIRA by implementing structured workflows and documentation standards, reducing task assignment delays by 30% and boosting collaboration efficiency for a team of 8 developers.
- Engineered SQL-based ETL processes to extract, transform, and load 1.5TB of financial data from heterogeneous sources, enabling high-speed analysis with 99.8% data integrity across 50+ datasets.
- Designed interactive Tableau dashboards to monitor 40+ financial KPIs, enhancing stakeholder decision-making by 20% and reducing quarterly review times by 15 hours for 10+ executives.

AI-Powered Unified Answer Platform: Python, SQL, REST APIs, AWS, Splunk

- Architected an AI-driven search platform with Python, automating manual data retrieval for 500+ queries daily, cutting processing time by 50% (from 10 minutes to 5 minutes per query) and improving operational efficiency.
- Integrated NLP algorithms with Splunk and REST APIs, enhancing search precision by 50% and enabling operations teams to resolve 200+ incidents monthly with 30% less effort.
- Deployed a serverless solution on AWS using Lambda, API Gateway, and CloudWatch, processing 1M+ requests monthly with 99.9% uptime and reducing infrastructure costs by 25% ($10K/year savings).

Serverless Task Tracker Web App: React, AWS (Lambda, API Gateway, DynamoDB, S3, CloudFront, CloudWatch), JS, HTML5, CSS, Git

- Developed a serverless task management web app with React, serving 1,200+ users across 5 organizations, delivering a scalable platform with zero server maintenance and 40% lower operational costs than traditional setups.
- Configured AWS Lambda and API Gateway for backend APIs, handling 18,000+ daily requests with sub-200ms response times, while leveraging DynamoDB to store 600GB of task data with 50% faster query performance (from 1s to 0.5s).
- Hosted React front-end on S3 with CloudFront CDN, accelerating page loads by 45% (from 2s to 1.1s) for users in 6 global regions, enhancing accessibility for remote teams.
- Utilized AWS CloudWatch to monitor 60+ Lambda functions, optimizing execution to slash costs by 35% ($5K/month saved) and ensuring 99.95% reliability during peak usage.
- Crafted a responsive UI with JavaScript, HTML5, and CSS, boosting task update efficiency by 60% (from 5min to 2min per update) for 350+ active users, while managing 25+ features across 20 JIRA-tracked Agile sprints using Git.

Predictive Analytics for Market Trends: Python, SQL, Power BI, Tableau, Machine Learning

- Built predictive models with Scikit-learn (regression, time-series forecasting), analyzing 2TB of energy market data to forecast trends with 25% higher accuracy (from 70% to 87.5%), supporting strategic planning for 5 energy firms.
- Constructed an Amazon Redshift data warehouse, integrating 10+ Power BI and Tableau dashboards to visualize model outputs, reducing insight delivery time by 40% (from 5 days to 3 days) for 15+ stakeholders.
- Executed SQL queries to aggregate and transform raw market data from 20+ sources into structured datasets, processing 500GB weekly with 99% consistency for machine learning pipelines.
- Developed Python scripts for feature engineering, enhancing model inputs by extracting 50+ predictive features, which improved forecast precision by 20% and cut training time by 15% (from 2h to 1.7h).